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CSIT314

Software Development

Methodologies

Rational Unified Process (RUP)



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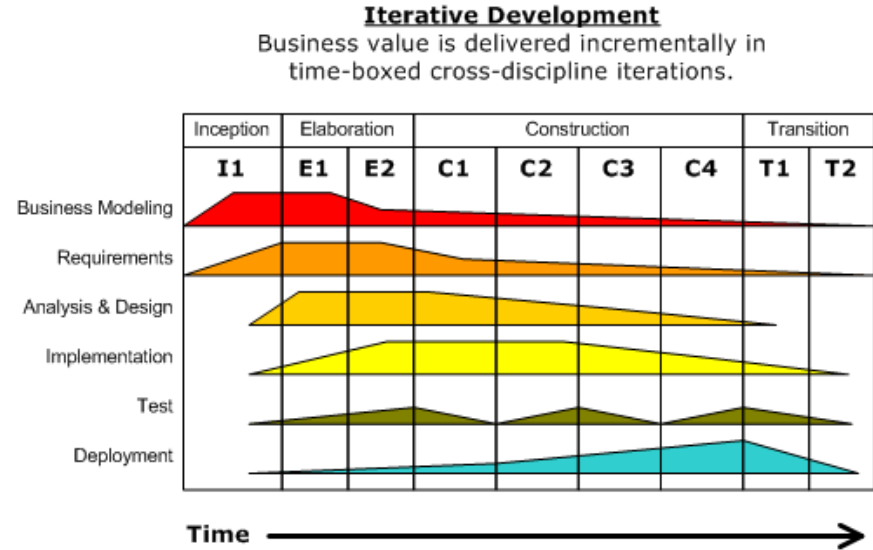
RUP

- Six principles to follow when designing any software project
 - Develop iteratively
 - Manage requirements: keep in mind the requirements set by customers and will be changed by users
 - Use components: test individual components and reusability
 - Model visually: use diagrams to represent all major components, users, and their interaction (UML)
 - Verify quality: always make testing a major part of the project at any point of time
 - Control changes: may be created by many teams, sometimes in various locations, different platforms may be used, etc.
 - make sure that changes made to a system are synchronized and verified constantly.



Rational Unified Process (RUP)

- Two dimensions
 - Horizontal axis represents time and shows the lifecycle aspects of the process as it unfolds
 - plan the scope and acceptance criteria
 - evaluate the resources and create actionable baseline
 - create, write, collaborate and test your software
 - releases the project
 - more later
 - Vertical axis represents core process workflows, which group activities logically by nature



1. Life-Cycle Objective Milestone
2. Life-Cycle Architecture Milestone
3. Initial Operational Capability Milestone
4. Product Release



A model of RUP

- A process describes who is doing what, how and when. The RUP is represented using five primary elements
 - Roles: who
 - Activities: how (also known as tasks)
 - Artifacts: what (also known as work products)
 - Workflows: when
 - Disciplines: container (configuration and change management, project management and environment)



RUP building blocks

- Roles (who) – A Role defines a set of related skills, competencies and responsibilities.
- Work Products (what) – A Work Product represents something resulting from a task, including all the documents and models produced while working through the process.
- Tasks (how) – A Task describes a unit of work assigned to a Role that provides a meaningful result.



Start at Inception

- Establish the project's software scope and acceptance criteria (requirements)
- Establish the main use-case scenarios that define the core functionality of the project.
- Evaluate alternative architectures; create the beginnings of your preferred candidate architecture.
- Estimate the overall cost and schedule for the project.
- Estimate risks
- Produce the “business case” – the business justification and benefit from having the functionality provided.
- Create a work plan for the following elaboration phase



Inception activities

- Formulate the scope of the project: capture the context and the most important requirements and constraints
- Plan and prepare a business case
- Evaluate alternatives:
 - risk management, staffing, project plan, and trade-offs among cost, schedule, and profitability
- Synthesize a candidate architecture



Inception outcomes

- A vision document
 - the core project's requirements, key features, main constraints
- List all use cases and actors already identified
- An initial project glossary (features)
- An initial business case, which includes the following
 - Business context
 - Success criteria (revenue projection, market recognition, and so on)
 - Financial forecast
- An initial risk assessment
- An initial project plan, which shows the phases and iterations

(example later)



Vision document

- What problem are we trying to solve? (Problem Statement)
 - describe the problem, who it affects, how it affects them, and what type of solution would ease the pain (solution can ...)
- Who are the stakeholders
 - persons or organizations. They may be affected by it either directly or indirectly
- Who are the users
- What are their respective needs
- What are the product features
- What are the functional requirements (Use Cases)
- What are the non-functional requirements
- What are the design constraints?



Identify features

- Example:
 - Record statistics
 - Time, Software defects and Source code size
 - Reporting
 - Personal reports and Team reports
 - Viewing
 - Statistics



Sketch high level use cases

- Software Engineer :
 - Create project, Enter data, Count items
 - Use on line help when needed, Create report
- Process administrator:
 - Configure system, Configure project
 - Install tools



Initial Project Plan

- “Iteration plan” for rest of inception phase
 - Artefact, who responsible, when due
- Example:
 - Plan done by 10th Oct
 - Environment set up ...
 - Vision complete by ...
 - Supplementary requirements
 - Initial use cases by ..
 - Risk list
 - Test plan
 - Finish “Inception Iteration”



Inception evaluation

- At the end of the inception phase is the first major project milestone. The evaluation criteria for the inception phase are:
 - Stakeholder concurrence on scope definition and cost and schedule estimates
 - Requirements understanding evident through use cases
 - Credibility of cost & schedule estimates, priorities, risks and development process
 - Depth and breadth of prototype architecture
 - Actual vs planned expenditure



Elaboration

- Requirements (resources) continue to be refined and additional use cases will be identified.
- Elaboration objectives:
 - Define, validate, and baseline the system architecture
 - Baseline the vision
 - Baseline a plan for the construction phase
 - Demonstrate that the baseline architecture will support this vision for a reasonable cost in a reasonable time



Elaboration activities

- The vision is elaborated and fully articulated.
- Develop solid understanding the vision of critical use cases.
- The process, the infrastructure, and the development environment are elaborated.
- Put in place development environment, tools, test automation systems, etc.
- Elaborate the system architecture:
 - Select subsystems that can be purchased
 - Integrate and assess selected components against primary use cases



Elaboration outcomes

- A use-case model (at least 80% complete) in which all use cases have been identified in the use-case model survey, all actors have been identified
 - Most use-case descriptions have been developed
- Supplementary requirements that capture the non-functional requirements
- A software architecture description
- An executable architectural prototype (database design)
- A revised risk list and a revised business case
- A development plan for the overall project
 - major and minor iterations and related deliverables
- A preliminary user manual (optional)



Elaboration evaluation

- Is the product vision stable?
- Is the architecture stable?
- Does the executable demo show that major risk elements have been addressed and resolved?
- Is the construction plan sufficiently detailed?
- Do all stakeholders agree that the current vision can be achieved with the current development plans?



Construction

- Develop remaining components and application features
 - coding, testing, and delivering iterated versions
 - cannot escape project creep
- Integrate all parts into the product
- Test all features thoroughly
- Achieve useful versions (alpha, beta, and other test releases) as rapidly as practical



Construction objectives

- Achieving useful versions as quickly as possible, or system requirement
- Minimized development costs by optimizing resource usage
 - avoiding unnecessary changes, scrap and rework
- Achieving adequate quality as rapidly as is practical



Construction activities

- Complete development and testing against the defined evaluation criteria
- Resource management, resource control, and process optimization
- Assessment of product releases against acceptance criteria for the vision



Construction outcome

- A product ready to put in the hands of its end users. It should consists of the following:
 - The software product integrated on the adequate platforms
 - The user manuals
 - A description of the current release



Construction evaluation

- Decide whether the software, the sites and the users are ready to become operational without exposing to high risks
 - Is the product stable and ready to be exposed to the user environment?
 - Are the stakeholders ready for the product transition into the user environment?
 - Are the actual resource expenditures still acceptable relative to the projected expenditures?



Iterations

- Plan (started during Inception and refined in Elaboration) will have defined a series of iterations
 - Each Iteration
 - List of use-cases that it incorporates
 - Time schedule (time boxing)
 - Leads to a “build” that can be release to others



Iteration releases – a chance to replan

- With each iteration release, you revise your plan
 - Maybe you are lagging
 - need to reduce the scope
 - move some features from next phase to later phase
 - Maybe you have negative feedback from the users/customers
 - Re-analysis, re-design, re-work



Transition

- Move the software product to the users
- Phase starts when system is mature enough to be deployed in the end-user domain
 - Usable subset of the system has been completed to an acceptable level of quality
 - User documentation is available
- Phase includes:
 - Beta testing to validate the new system against users' expectations
 - Parallel operation with any existing system that the project is replacing
 - Training of users and maintainers
 - Rollout of the product to the marketing, distribution, and sales team



Transition objectives

- Achieve user self-supportability
- Achieve stakeholder concurrence that deployment baselines are complete and consistent with the evaluation criteria of the vision
- Achieve final produce baseline as rapidly and cost effectively as practical



Transition activities

- Deployment-specific engineering, i.e., commercial packaging and production, sales rollout and etc.
- Tuning activities, including bug fixing and enhancement for performance and usability
- Assessing the deployment baselines against the vision and the acceptance criteria for the product



Transition evaluation

- You decide whether the objectives were met and whether you should start another development cycle. The primary evaluation criteria are:
 - Is the user satisfied?
 - Are the actual resources expenditures versus planned expenditures still acceptable?

